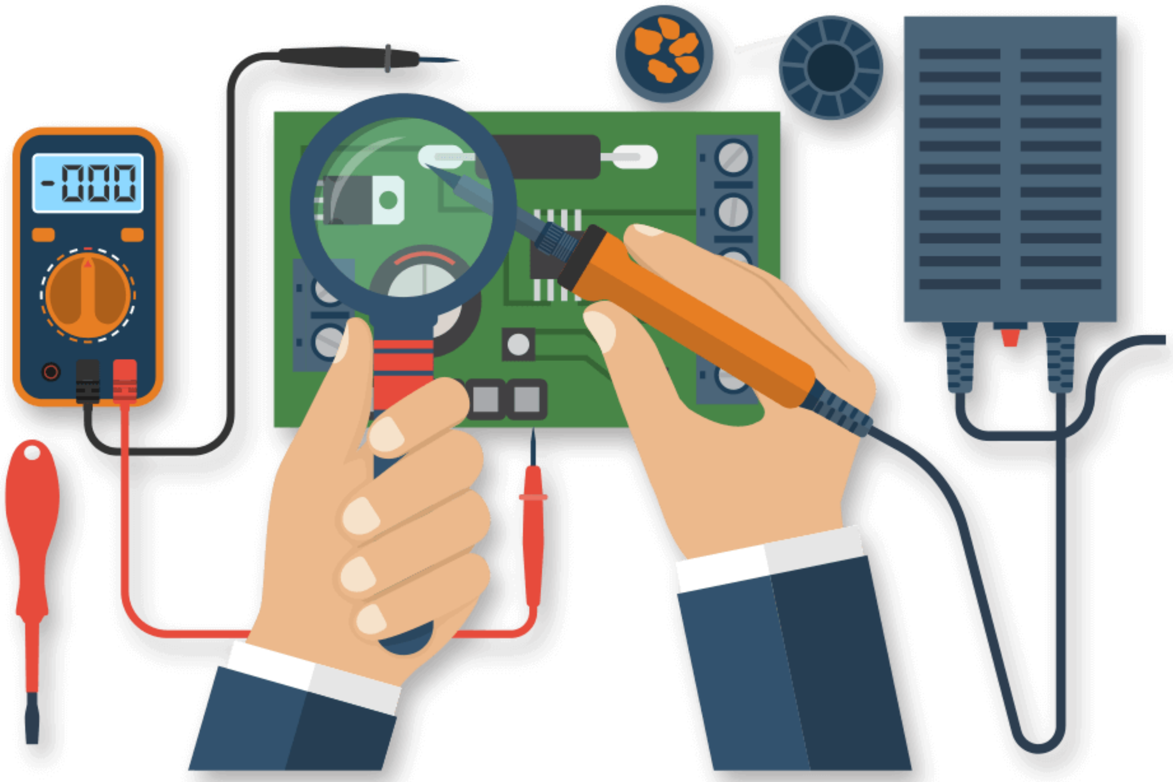
Computer System Architecture



Day 1: Basics of Electricity

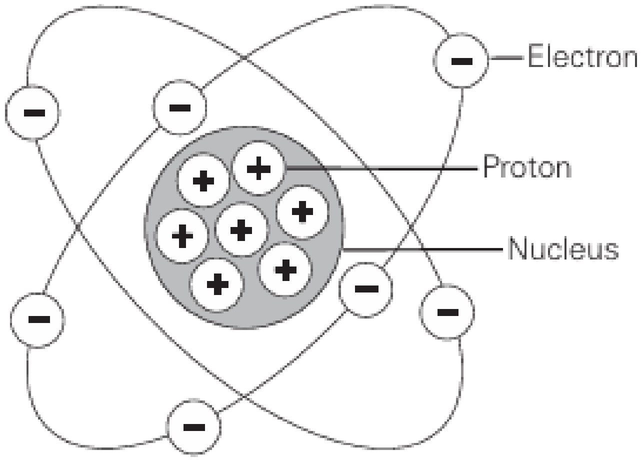
Supun Chandana Gajendrasinghe
Department of Network and Security
supun.g@nsbm.ac.lk

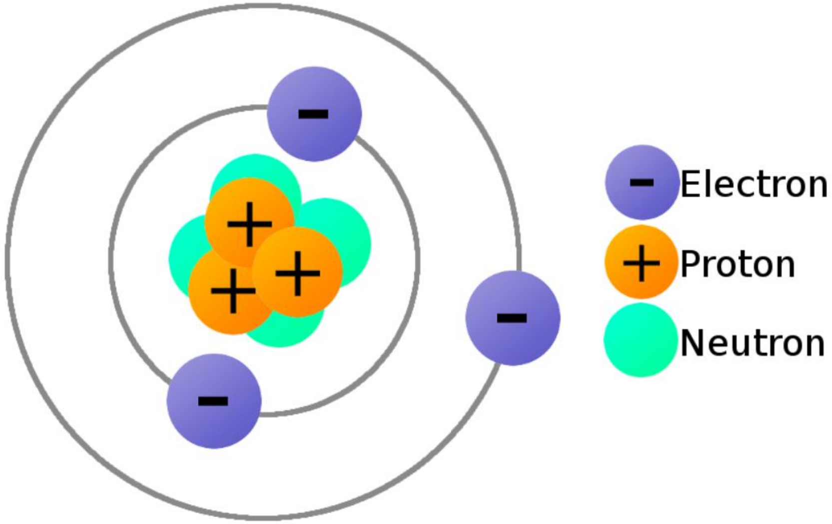
Outline

- Electron Theory
- Conductors, Semiconductors, Insulators
- Electric Charges
- Current
- Voltage
- Resistance

Electron Theory

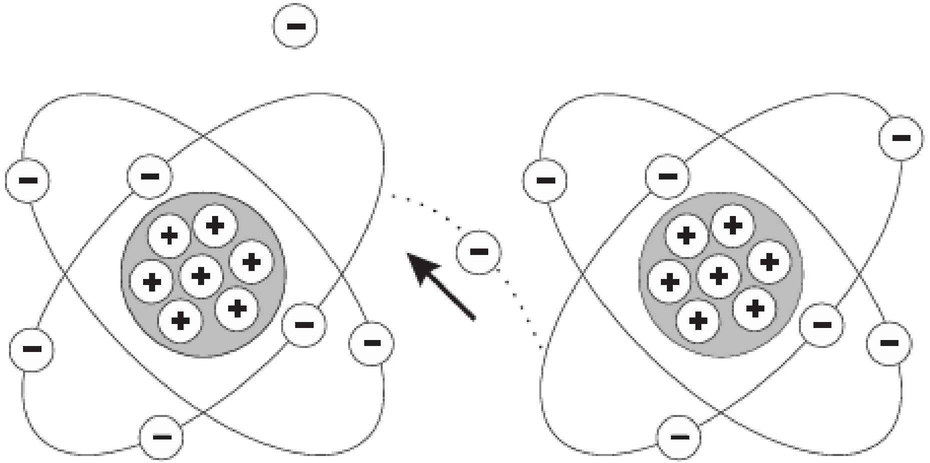
Elements of an Atom

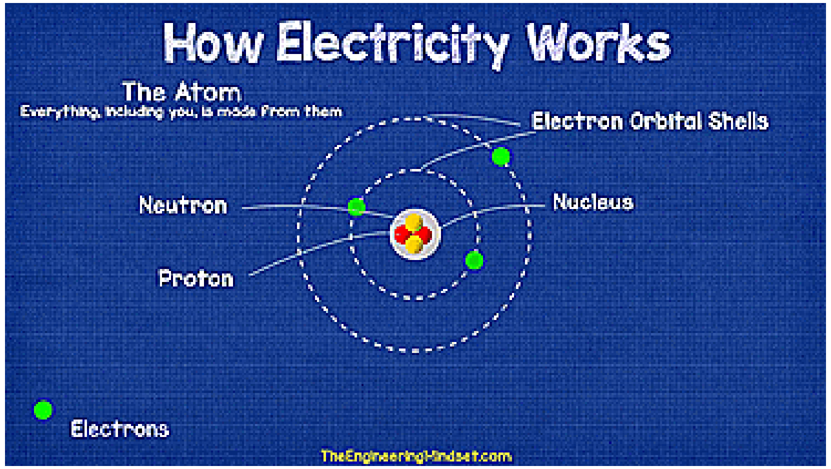




Electron Theory Cont`d…

Free Electrons

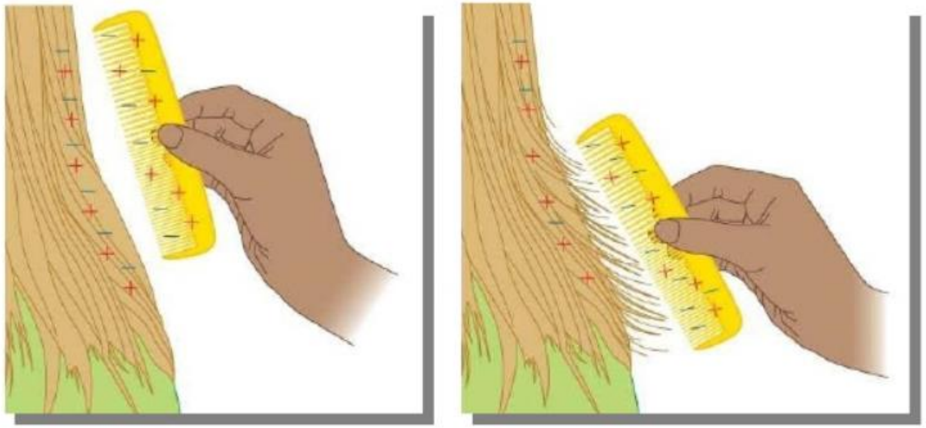




Electrons in the outer band can become free of their orbit by the application of some external force such as **movement through a magnetic field, friction, or chemical action**.

Electron Theory Cont`d…

Free Electrons



- Static Electricity by friction: as one object loses electrons, the other object gains them.
- An electric current is produced when **free electrons move from one atom to the next**.

Electron Theory Cont`d…

Antistatic Wrist Strap

To prevent the buildup of static electricity on their body, which can result in ESD.

Conductors, Semi-Conductors, Insulators

Conductors

- Materials that **permit many electrons to move freely** are called conductors.
- Copper, silver, aluminum, zinc, brass, and iron are considered good conductors.
- Copper is the most common Conductor.

Conductors, Semi-Conductors, Insulators Cont`d…

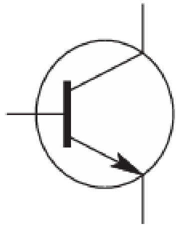
Insulators

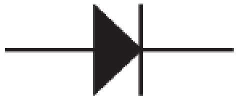
- Materials that **allow few free electrons** are called insulators.

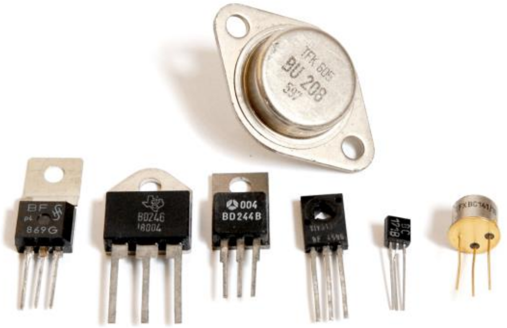
Conductors, Semi-Conductors, Insulators Cont`d…

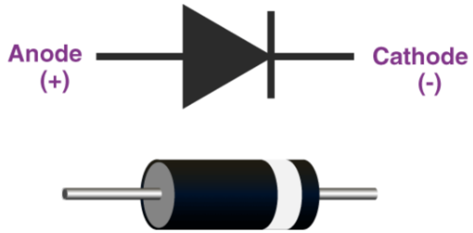
Semi - Conductors

- have characteristics of both conductors and insulators.
- Many semiconductor devices will act like a conductor when an external force is applied in one direction.
- When the external force is applied in the opposite direction, the semiconductor device will act like an insulator.
- Silicon is used for manufacturing semi conductor devices.


Transistor

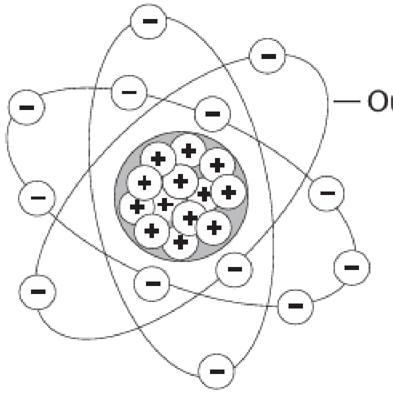

Diode





Electric Charges

Neutral State of an Atom

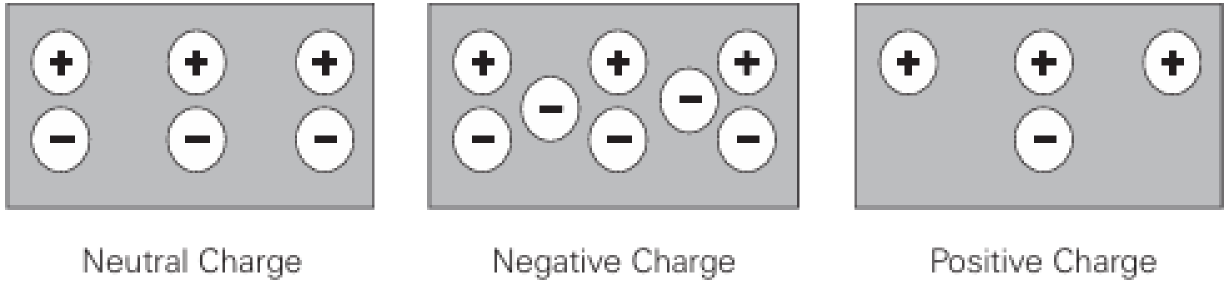


An atom with an **equal number of electrons and protons** is said to be electrically neutral.

E.g. Aluminum Atom has 13 Electrons and 13 Protons

Electric Charges Cont`d…

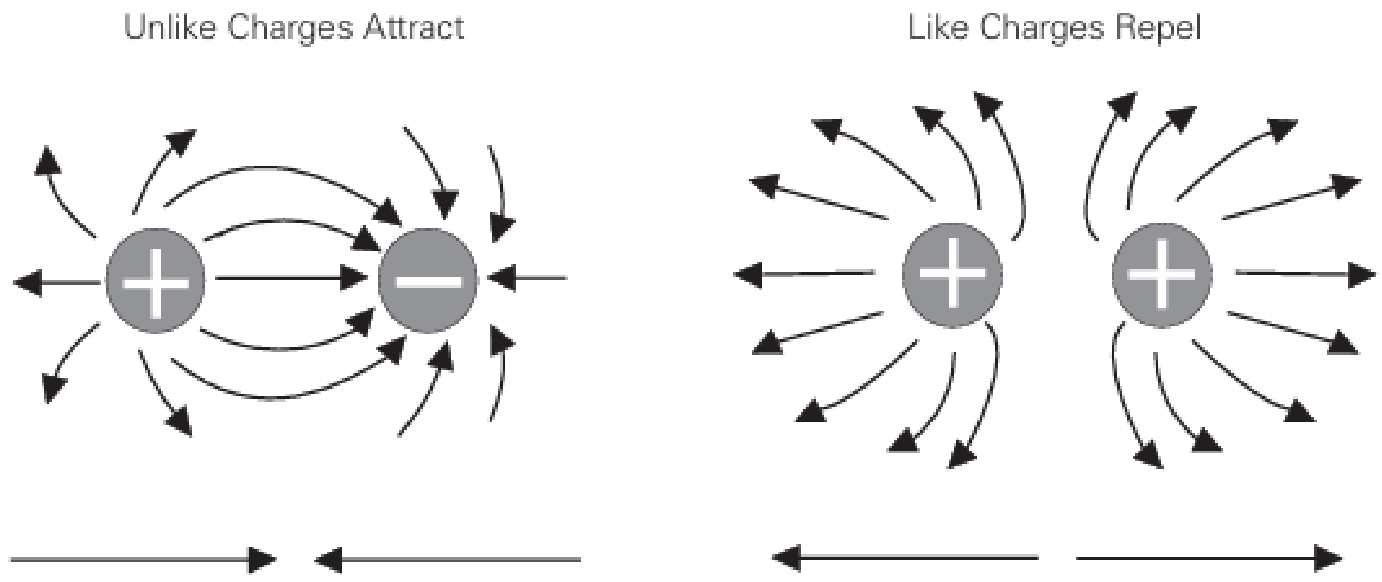
Positive and Negative Charges



A positive or negative charge is caused by an absence or excess of electrons. The **number of protons remains constant**.

Electric Charges Cont`d…

Attraction and Repulsion of Charges

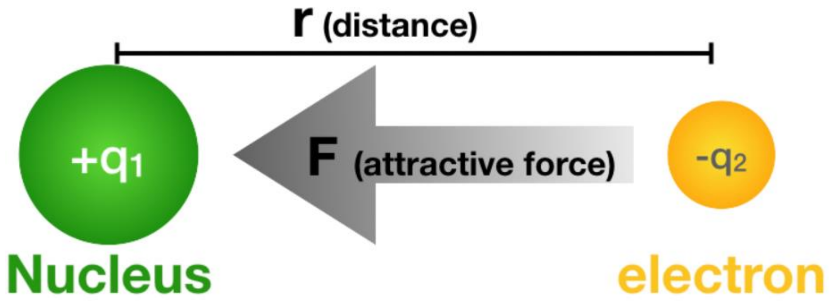


Electric Charges Cont`d…

Coulombs Law

$$F = k\frac{q_1 q_1}{r^2}$$

The force of attraction or repulsion depends on the strength of the charged bodies, and the distance between them



Current (I)

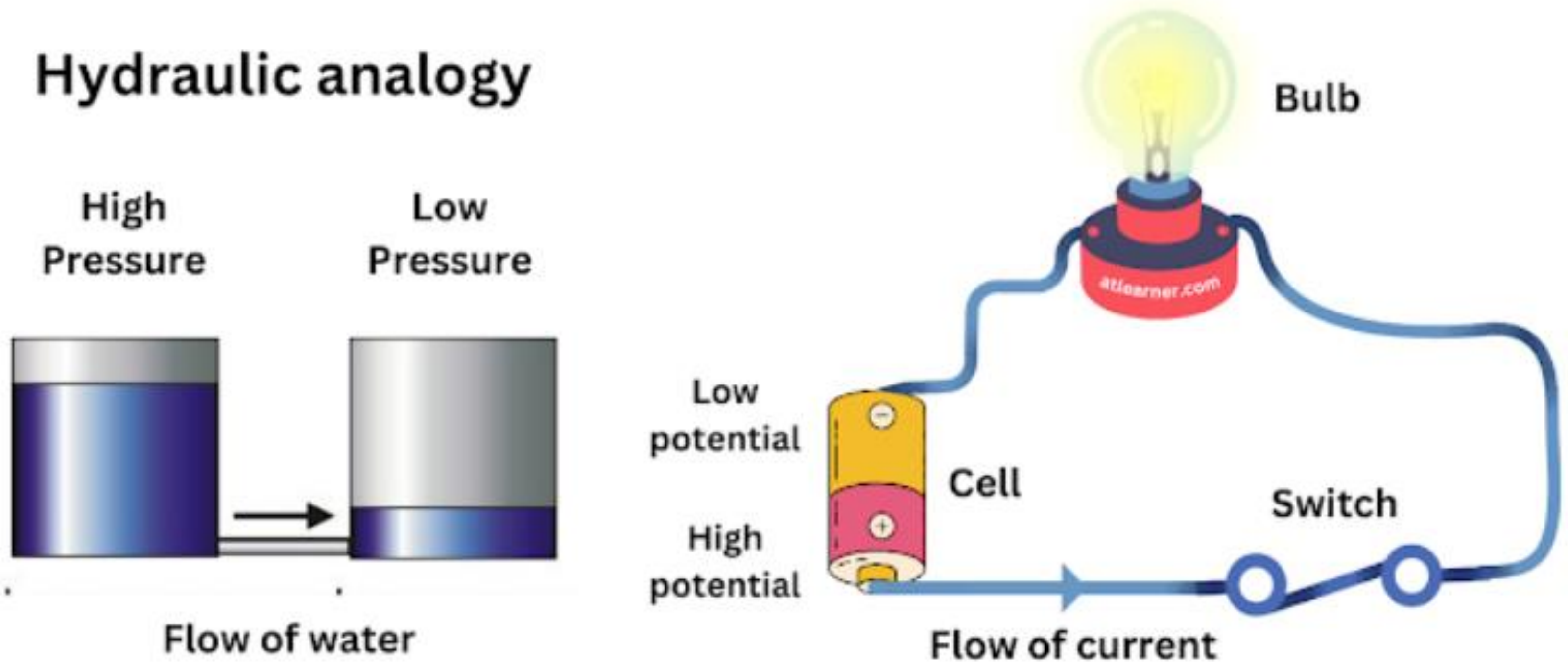
- Electricity is the flow of free electrons in a conductor from one atom to the next atom in the same general direction.

- **flow of electrons** is referred to as current.

- Current is determined by the **number of electrons that pass through a cross-section of a conductor in one second**.

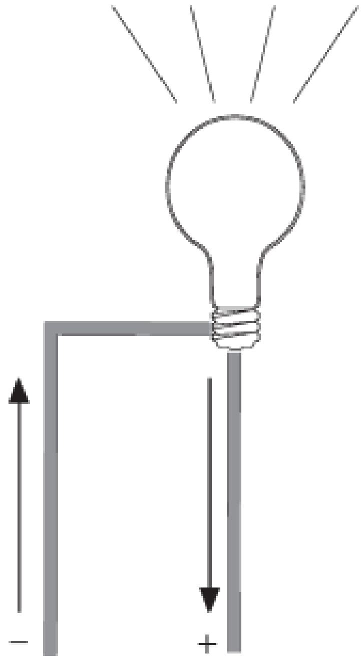
- Measured in Amperes (A).

Current (I) Cont`d…

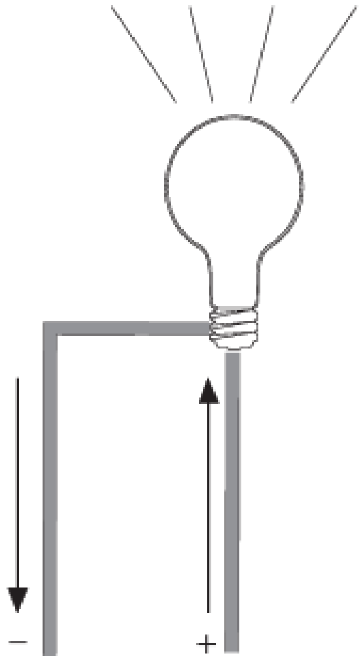


Current (I) Cont`d…

Direction of Current Flow



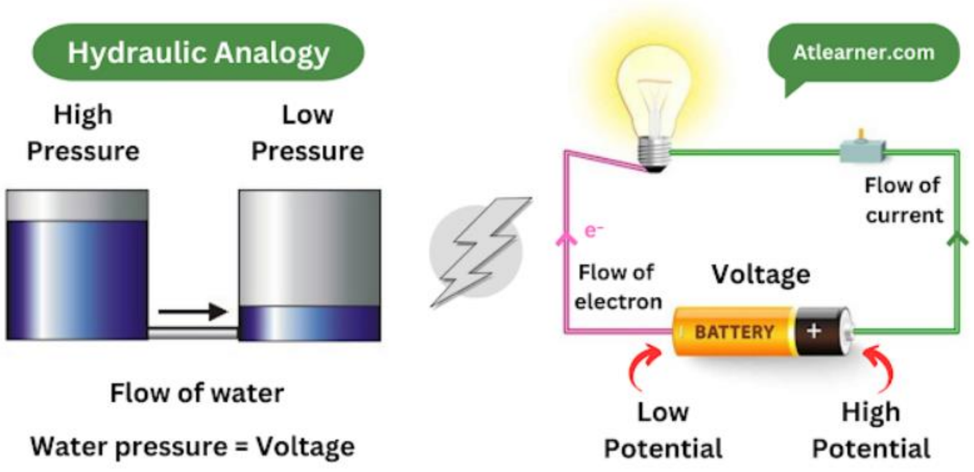
Electron Flow



Conventional
Current Flow

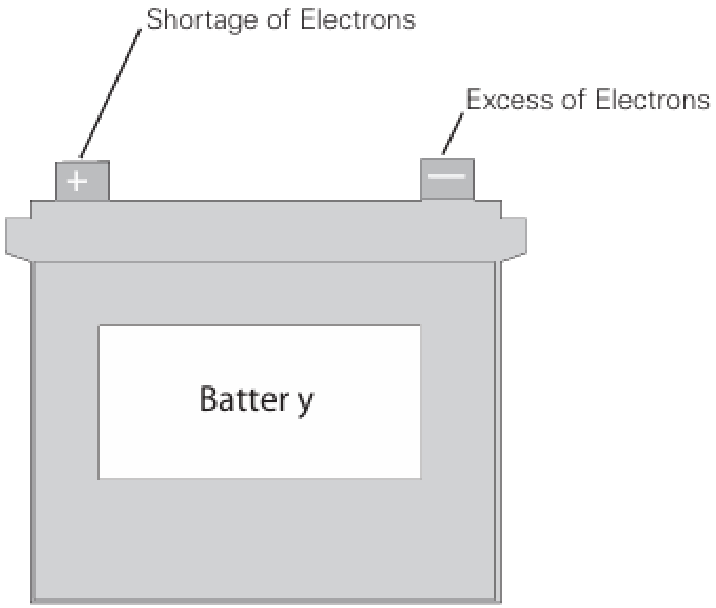
Voltage (V)

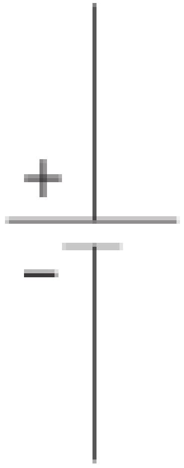
- Force that is applied to a conductor that causes electric current to flow.
- Measured in Volts (V)



Voltage (V) Cont`d…

Voltage Sources

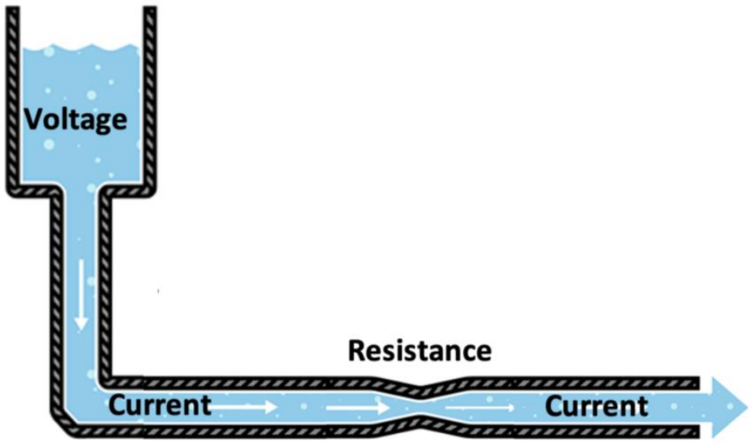


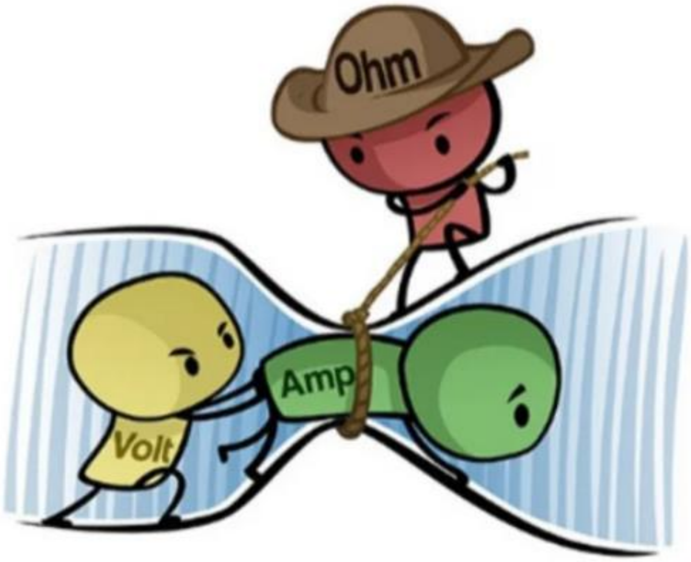


Symbol of Battery

Resistance (R)

- Opposition to current flow
- Measured in Ohms (Ω)





Thank you!

Questions